CURBCOUNT · PRODUCT REQUIREMENTS AND SPECIFICATIONS

CurbCount product requirements and specifications

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CBC-PRS-001	Version	0.7
Status	Draft	Date	2026-10-31
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/03-requirements.md @ 82b4850		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: restated as product requirements PR-001 to PR-021, each traced to a user requirement in CBC-URS-001; old R numbers kept in the Was column; document type PRS (was REQ, CBC-REQ-001); R9 now at risk below about -5 C air (STEM check S8); R7 and R12 split; mass 5.24 kg from CBC-CAL-001 sizing.py
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status figures from CBC-CAL-001 v0.3 after the design for construction (CBC-DDR-003); no status changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from CBC-CAL-001; design case bike lane 2.0 m to match R6; decisions from CBC-DDR-001 reflected; proposed restatements of R4 and R11 noted

0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CurbCount product requirements and specifications

What the counter must do, each requirement traced to a user need in CBC-URS-001. Targets are proposals for review with a co-design partner. Method: STANDARDS section 22. Status comes from the calculation note CBC-CAL-001 (v0.4 and `sizing.py`) and the STEM check CBC-STM-001; "met on paper" means met by calculation only. **No requirement is not met.** Five are at risk (PR-001, PR-002, PR-008, PR-010 and PR-016) and five cannot be shown at TRL 3 and are marked to verify at TRL 4 (PR-003, PR-007, PR-011, PR-013 and PR-017).

The design choices behind these requirements (thermal array, tracking on FieldNode's STM32WL with standard FieldNode power, 15-minute bins and three classes in a 10-byte record, jumper-only calibration, public notice) were decided by Amish on 2026-09-25 (CBC-DDR-001 and CBC-DDR-002). Version 0.7 restates the fifteen requirements R1 to R15 of CBC-REQ-001 v0.6 as product requirements PR-001 to PR-021 (the Was column keeps the old numbers) and traces each to a user requirement.

Design case: one counter on a 114 mm pole 0.45 m behind the curb, sensor window 4.3 m above the road, a 3 m sidewalk, a 2.0 m bike lane and one 3.5 m traffic lane.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
PR-001	UR-003	Performance	Count three classes in two directions	The counter shall count pedestrians, cyclists (including micromobility) and motor vehicles in the nearest lane, by direction along the street	6	min	counters (3 classes x 2 directions)	Analysis (CAL-001 C), TRL 3; test against manual counts, TRL 4	At risk: 4.2 frames for a point at road level, 3.2 for a warm hood 1.0 m up (STM S5), against 4 assumed needed; fix proposed to Amish (CBC-DEC-001 O3)	R1
PR-002	UR-004	Performance	Pedestrian and cyclist accuracy	The hourly count of pedestrians and cyclists shall be within 10 % of a manual count at flows up to 600 per hour per direction	10	max error	%	Analysis (CAL-001 C), TRL 3; field comparison with manual counts, TRL 4	At risk: random merging alone loses 6.5 %; groups add more	R2
PR-003	UR-004	Performance	Motor vehicle accuracy	The hourly count of motor vehicles in the nearest lane shall be within 15 % of a manual count	15	max error	%	Analysis (CAL-001 C), TRL 3; field comparison, TRL 4	To verify at TRL 4: 4.2 frames per car, 3.4 px smear	R3
PR-004	UR-006	Safety	Sensor cannot resolve a face or a plate	No image, audio or identifier shall leave the device and frames shall be held only in RAM; the sensor shall give at most 8 px/m at head height for a 1.75 m person, below the IEC	8	max	px/m	Analysis (CAL-001 B), TRL 3; design review and firmware audit, TRL 4	Met on paper: 7.6 px/m (8.5 for a 2.0 m person) for passers-by; an installer within 0.5 m is above the detection level	R4

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				62676-4 detection level of 25 px/m						
PR-005	UR-007	Software	Open 10-byte count record	The counter shall send 15-minute counts per class and direction in an open, documented record of six 12-bit counters and a status byte over LoRaWAN	10	max	bytes	Inspection (design review, CAL-001 G), TRL 3	Met by design	R5
PR-006	UR-005	Performance	Coverage from one pole	One counter shall cover the full width of a 3 m sidewalk, a 2 m bike lane and the nearest 3.5 m traffic lane	8.5	min	m across the street	Analysis (CAL-001 A), TRL 3; field check, TRL 4	Met on paper with the array turned 90 degrees: -4.45 to 8.34 m across	R6
PR-007	UR-009	Materials	Survive -20 to +50 degrees C	The counter shall survive air temperatures from -20 degrees C to +50 degrees C	70	span	K	Inspection (part ratings, STM S7, S8), TRL 3; thermal chamber, TRL 4	To verify at TRL 4: cell discharge rating and window film not yet measured	R7
PR-008	UR-009	Performance	Count in sun and shade at temperate sites	The counter shall count at temperate sites, on design days with air up to 20 degrees C; hot-climate sites use the radar variant (CBC-DDR-002)	20	max air	degrees C	Analysis (CAL-001 D, STM S3), TRL 3; seasonal field test, TRL 4	At risk: on the mild design day sunlit pavement hides pedestrians for 10.2 h with single frames, 3.2 h with track averaging; shaded 0 h. Summer days above	R7

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
									20 degrees C are outside the claim (the independent review put a 15 to 28 degrees C day at about 12 h blind on shaded pavement; not recomputed here)	
PR-009	UR-010	Electrical	Three days without sun	The counter shall keep counting for 3 days without sun	3	min	days	Analysis (CAL-001 E2), TRL 3; bench test, TRL 4	Met on paper: 6.22 days on one cell (4.36 at -20 degrees C, 3.49 with the end-of-life 80 % as well)	R8
PR-010	UR-010	Electrical	Energy neutral in winter	The panel shall store more energy per day than the counter uses at 1.5 peak sun hours with a 40 % street derating, while the cell is allowed to charge (air above about 0 degrees C, STM S8)	1.5	min	peak sun hours	Analysis (CAL-001 E3, STM S6 and S8), TRL 3; field log, TRL 4	At risk below about 0 degrees C air (cell may not charge below 0 degrees C; CBC-DEC-001 O1). Met on paper above it: 4.4 Wh/day stored against 2.47 Wh/day	R9
PR-011	UR-011	Physical architecture	IP65 enclosure and head	The enclosure and the sensor head shall meet IP65	65	min	IP rating	Inspection (design review, CAL-001 I), TRL 3; spray test, TRL 4	To verify at TRL 4	R1
PR-012	UR-002	Physical architecture	Mass on the pole	The counter shall have a	6	max	kg	Analysis (CAL-001 H5,	Met on paper: 5.24	R1

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				total mass of 6 kg or less				cad/ fidelity.json), TRL 3; weighing, TRL 4	kg with bought parts (CAD model 4.8 kg)	
PR-013	UR-001	Usability	Install in 60 minutes by two people	Two trained people shall install the counter in 60 minutes or less	60	max	min	Analysis (CAL-001 H6, BLD-001), TRL 3; timed trial, TRL 4	To verify at TRL 4: 59 min estimate for the task list in CAL-001 H6; it leaves out platform set-up, traffic control and teardown, which the independent review put at 75 to 90 min in all	R1
PR-014	UR-001	Physical architecture	No drilling, welding or pole wiring	The counter shall fit with band clamps and a pole-top sleeve, with no drilling, welding or connection to the pole's wiring	0	max	pole modifications	Inspection (BLD-001, BOM lines 5 to 7), TRL 3	Met by design	R1
PR-015	UR-015	Physical architecture	Withstand 35 m/s gusts	The mounting shall withstand 35 m/s gusts with a safety factor of at least 1.5 on the pole top, arm and head	35	min	m/s	Analysis (CAL-001 H1, H3), TRL 3; bracket calculation, TRL 4	Met on paper: panel 52 N, post factor 55, twist factor 3.3	R1
PR-016	UR-013	Packaging and transport	Parts cost within the value-engineering target	The priced BOM shall total 275 USD or less (a value-	275	max	USD	Analysis (BOM, CAL-001 J1), TRL 3	At risk: 274.00 USD is 1.00 USD (0.4 %) under the	R1

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				engineering target, not a limit)					target, less than the uncertainty in indicative prices	
PR-017	UR-011, UR-012	Materials	Five years outdoors, one cell change	The counter shall last 5 years outdoors with one battery change, using UV-stable materials	5	min	years	Design review (CAL-001 I, STM S18 and S19), TRL 3; ageing and field trial, TRL 4	To verify at TRL 4: window film grade and joint coatings not yet specified	R1
PR-018	UR-014	Electrical	Radio within licence-free limits	Uplinks shall stay within the EU868 1 % duty cycle and the US915 400 ms dwell limit	1	max	% duty	Analysis (CAL-001 G), TRL 3	Met on paper: 0.82 s/h at SF9 (limit 36 s/h); 371 ms at US915 DR0	R1
PR-019	UR-016	Usability	Public notice plate	A notice plate on the pole shall state what is counted, that no image is stored or sent, and the repository link	1	min	plate	Inspection (BOM line 12, BLD-001), TRL 3	Met by design (CBC-DDR-001 D7)	R4
PR-020	UR-008	Usability	Aim check on site	A crew shall be able to check the aim and confirm an uplink with a laptop or phone, using the calibration jumper and no special software	10	max	min	Analysis (CAL-001 H6), TRL 3; demonstration, TRL 4	Met by design: 8 min estimate	R1
PR-021	UR-015	Safety	Panel and head retained by lanyards	The panel mount and the sensor head shall each be held by a lanyard so that a	2	min	lanyards	Inspection (BLD-001, BOM line 13), TRL 3	Met by design	R1

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				failed fixing does not drop a part onto the street						

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Changes from CBC-REQ-001 v0.6

- R7 is split into PR-007 (survive -20 to +50 degrees C) and PR-008 (count at temperate sites). R12 is split into PR-013 (install time), PR-014 (no drilling), PR-015 (wind), PR-020 (aim check) and PR-021 (lanyards).
- PR-010 (was R9) is now at risk for sites where the air stays below about 0 degrees C: a LiFePO4 cell must not be charged below 0 degrees C and FieldNode's lockout stops charging (STM S8). The fix changes the shared FieldNode core, so it is an open decision for Amish (CBC-DEC-001 O1).
- PR-019 records the public notice plate decided on 2026-09-25 as its own requirement.
- PR-012 reads 5.24 kg, the figure `sizing.py` now prints for the model with fixings as individual parts; CBC-REQ-001 v0.6 said 5.26 kg.

Assumptions

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Walking speed	1.0 to 1.5 m/s	Typical design range (estimate)
Cyclist speed	up to 25 km/h	Typical urban maximum (estimate)
Vehicle speed in the nearest lane	up to 50 km/h	Common urban limit (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Street shading and dust derating	40 %	Estimate, to be checked on real poles
Temperate design day	Air 10 to 20 degrees C	CBC-CAL-001 section D